

Git & GitHub Full Manual Setup & Rules Guide (English)

Complete Standard Operating Procedure (SOP) to Initialize, Configure, and Push Any Project to GitHub

MASTER GIT RULE

TERMINAL & CMD READY

MULTI-ACCOUNT SAFE

GITHUB DESKTOP

ERROR RESOLUTION

1. Introduction & Security Philosophy

This document establishes the official SOP for version controlling and publishing any software solution (such as `PCAccess`) to GitHub from scratch. Following these exact steps guarantees safe initialization, zero credentials leakage, proper exclusion of build artifacts, and smooth pushing via Command Prompt, PowerShell, or GitHub Desktop.

💡 **Security Authentication Rule:** Since August 2021, GitHub permanently deprecated direct password authentication for Git operations. All pushes require either **Browser OAuth (Git Credential Manager)** or a **Personal Access Token (PAT)**.

2. Pre-Staging: .gitignore Architecture (Mandatory First Step)

Before staging any .NET / Visual Studio solution into Git, a robust `.gitignore` must exist at the workspace root to prevent tracking large binary builds, caches, user-specific settings, and temporary files.

```
# Essential .gitignore rules for Visual Studio solutions: [Bb]in/ [Oo]bj/ packages/ .vs/ *.user  
*.suo *.bak
```

3. End-to-End Command Flow (Terminal / CMD / PowerShell)

1 Navigate to the Project Root

Open Command Prompt or PowerShell and switch to the repository folder:

```
cd /d E:\Git\PCAccess
```

2 Initialize Git Repository

```
git init
```

This creates the hidden `.git/` directory holding the repository database and tracking metadata.

3 Configure Project-Specific Identity (Local Isolation)

Avoid changing the workstation's global Git profile. Always assign the company / client identity locally to the repo:

```
git config user.name "Visibit Technologies" git config user.email "visibittechnologies@gmail.com"
```

4 Standardize the Default Branch to 'main'

```
git branch -M main
```

5 Stage and Commit All Solution Files

```
# Stage all tracked and new files (excluding those in .gitignore): git add . # Create the snapshot  
with a meaningful message: git commit -m "Initial commit: Project full codebase with documentation"
```

6 Create Repository on GitHub

1. Navigate to: **<https://github.com/new>** in your web browser.
2. Enter the Repository Name (e.g., `PCAccess-` or `PCAccess`). (Notice exact spelling and trailing characters).
3. Choose Public or Private, leave "Add README" unchecked, and click "**Create repository**".

7 Link Remote Origin (★ The Multi-Account Golden Rule)

If Windows Credential Manager has another GitHub account cached (e.g., a personal developer account), a standard remote link will fail with **HTTP 403 Forbidden**.

The Proven Resolution: Explicitly prefix the target GitHub username in the remote URL:

```
# Format: https://<username>@github.com/<username>/<repo>.git git remote add origin  
https://visibittechnologies@github.com/visibittechnologies/PCAccess-.git
```

To update an existing remote URL:

```
git remote set-url origin https://visibittechnologies@github.com/visibittechnologies/PCAccess-.git
```

8 Push to GitHub Remote

```
git push -u origin main
```

What to expect: Git Credential Manager will invoke a browser window. If you are already signed into GitHub as `visibittechnologies` , simply click the green "**Authorize**" button. Your entire codebase will push cleanly.

4. Troubleshooting Matrix (Common Errors & Solutions)

Error Message	Root Cause	Exact Solution
remote: Repository not found	Repository not created yet on GitHub, or character mismatch (e.g. <code>PCAccess</code> vs <code>PCAccess-</code>).	Verify browser URL and re-point remote: <code>git remote set-url origin <exact_url></code>
Permission denied to <user> (HTTP 403)	Windows cached credentials for another account.	Inject target username into URL: <code>https://visibittechnologies@github.com/...</code>
Password authentication was removed	Account password was entered into terminal.	Use browser OAuth prompt or Personal Access Token (PAT) with <code>repo</code> scope.
fatal: not a git repository	Command run in directory without Git initialized.	Run <code>git init</code> in the project directory.

5. GitHub Desktop Integration Rules

Rule A: On This Same PC (Where Code Already Exists)

- 1. Launch GitHub Desktop.
- 2. Go to top menu: **File → Add Local Repository...** (Shortcut: `Ctrl + O`).
- 3. Select directory: `E:\Git\PCAccess` .
- 4. Click **Add Repository**.

Important: Never select "Clone" for this directory on this PC, as it will throw *"Folder already exists"*.

Rule B: On a Fresh PC or for Team Members

- 1. Launch GitHub Desktop and log in.
- 2. Select: **File → Clone Repository...**
- 3. Pick `PCAccess-` from the repository list and choose an empty destination folder.
- 4. Click **Clone** to fetch all source code, agents, and release packages.

6. Daily Git Operations Cheat Sheet

Operation	Command
Inspect working tree status	<code>git status</code>
Stage all modifications	<code>git add .</code>
Record staged changes	<code>git commit -m "Description of changes"</code>
Upload commits to GitHub	<code>git push</code>
Fetch and merge remote updates	<code>git pull origin main</code>
Review recent commit history	<code>git log --oneline -n 5</code>